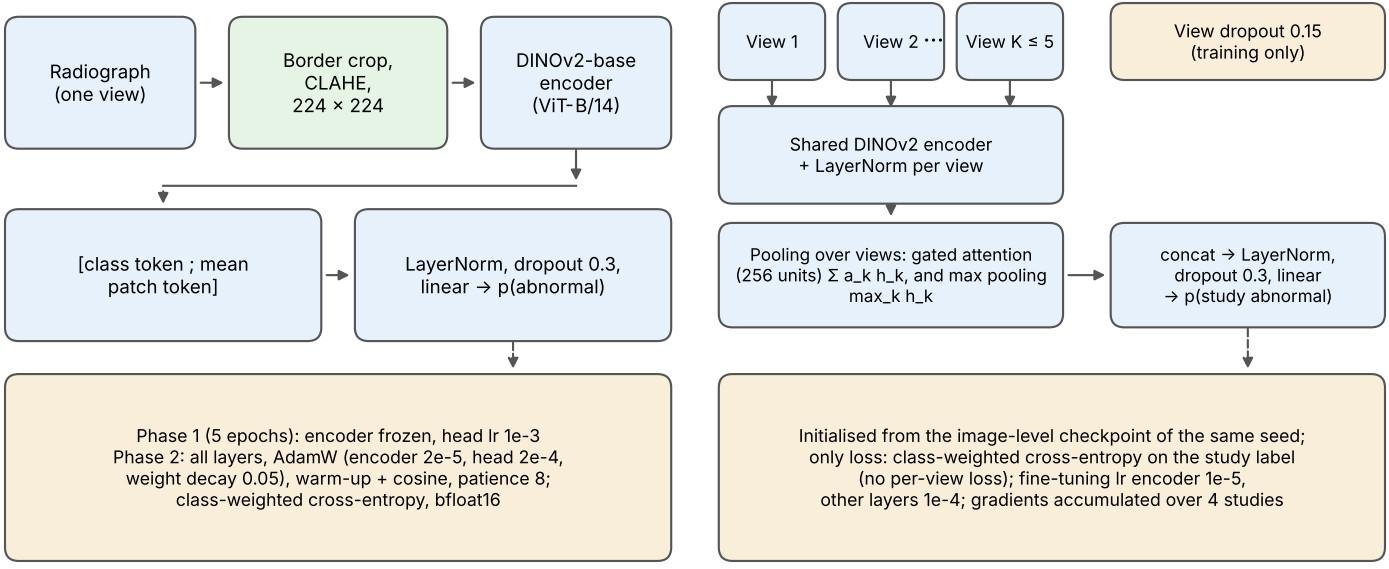
(a) Image-level DINOv2 model
Radiograph (one view)
Border crop, CLAHE, 224 × 224
DINOv2-base encoder (ViT-B/14)
[class token ; mean patch token]
LayerNorm, dropout 0.3, linear → p(abnormal)
Phase 1 (5 epochs): encoder frozen, head lr 1e-3
Phase 2: all layers, AdamW (encoder 2e-5, head 2e-4, weight decay 0.05), warm-up + cosine, patience 8; class-weighted cross-entropy, bfloat16
(b) Study-level multiple-instance model
View 1
View 2 ⋯
View K ≤ 5
View dropout 0.15 (training only)
Shared DINOv2 encoder + LayerNorm per view
Pooling over views: gated attention (256 units) Σ a_k h_k, and max pooling max_k h_k
concat → LayerNorm, dropout 0.3, linear → p(study abnormal)
Initialised from the image-level checkpoint of the same seed; only loss: class-weighted cross-entropy on the study label (no per-view loss); fine-tuning lr encoder 1e-5, other layers 1e-4; gradients accumulated over 4 studies